Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the validation accuracy during training.

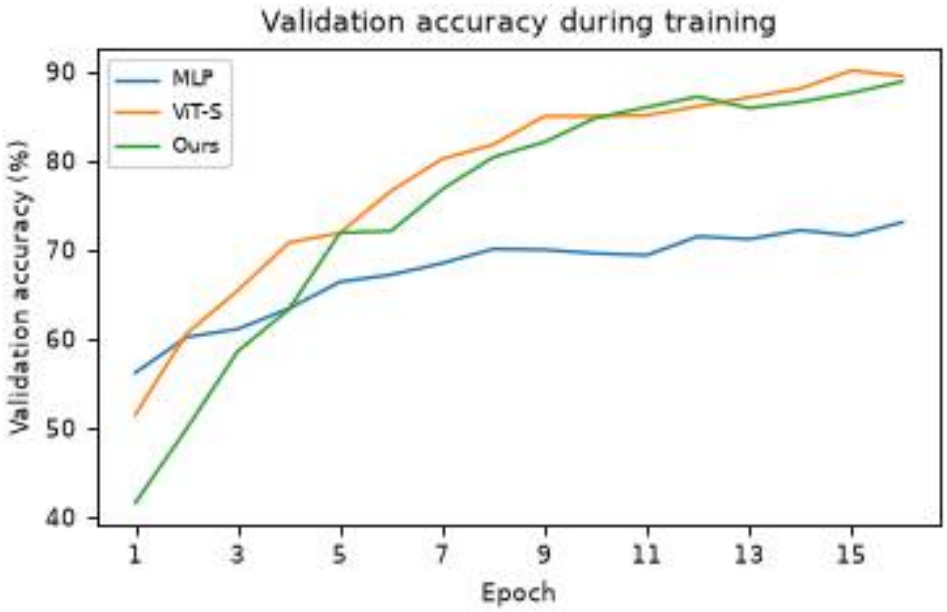


Figure 1: Validation accuracy during training for the models in this experiment.

From epoch 2 to epoch 12, the validation accuracy of Ours improves by a relative 74.7%. ViT-S reaches 88.4% validation accuracy at epoch 16. ViT-S reaches 63.4% validation accuracy at epoch 15. ViT-S reaches 86.0% validation accuracy at epoch 11 (Figure 1).

Figure 2 shows the validation accuracy by dataset.

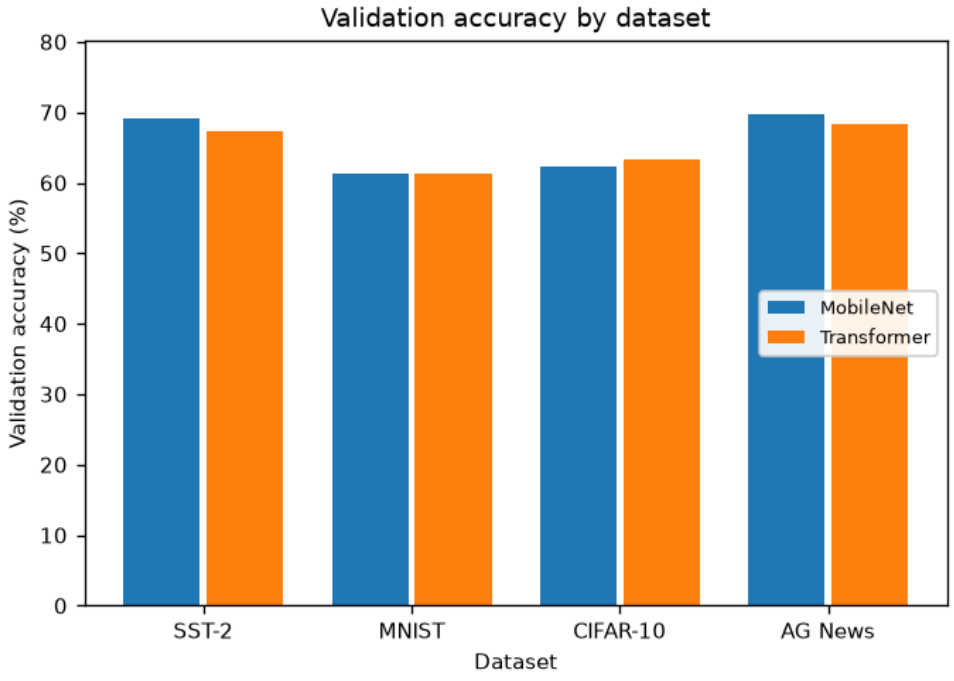


Figure 2: Validation accuracy by dataset for the models in this experiment.

Transformer achieves 67.4% validation accuracy on SST-2 (Figure 2). On AG News, MobileNet is 5.3 points better than Transformer (Figure 2).

Figure 3 shows the validation accuracy by dataset.

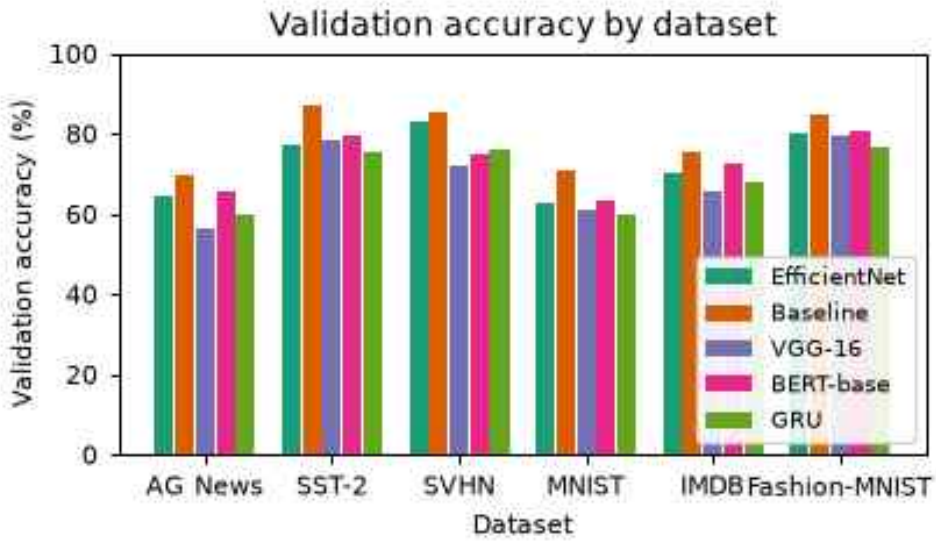


Figure 3: Validation accuracy by dataset for the models in this experiment.

EfficientNet reaches about 80% validation accuracy on SVHN. As shown in Figure 3, on IMDB, EfficientNet obtains a validation accuracy of 65.3%. As shown in Figure 3, Baseline reaches 55.4% validation accuracy on IMDB.

3 Discussion

Random Forest reaches 64.0% accuracy on QQP. We note that the training was stable for all models.